Causal Inference

- moving beyond simple connections



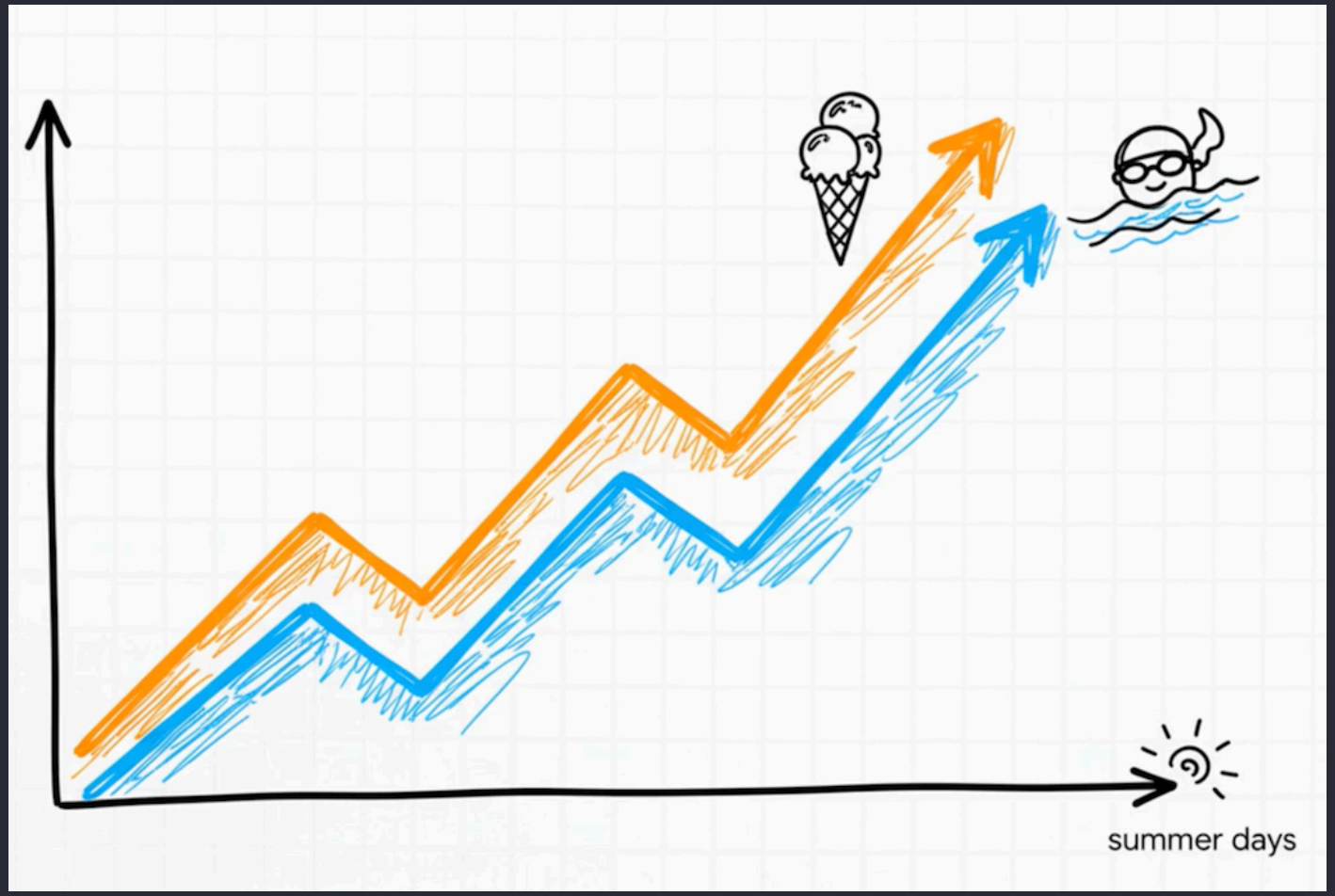
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Watch: Correlation and Causation

- Short primer on why correlation alone never proves causation
- Correlation and causation

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The Correlation Trap



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The Correlation Trap

- **The Observation**: On days when ice cream sales spike, the number of drownings goes up
- **The Logical Error**: Thinking that eating ice cream causes drowning
- **The Fix**: Using causal inference to dig deeper and find hidden forces

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Ladder of Causation

- cause-and-effect thinking
  - visualized as
    - three-rung ladder



Rung 1: Association

- **Nature**: Pure observation and finding patterns
- **Question**: "How are these things related?"
- **Example**: Noticing ice cream sales and drownings show up together
- **Utility**: Great for prediction, but it doesn't tell you *why*

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Rung 2: Intervention

- **Nature**: Active participation through experiments
- **Question**: "What happens if we act?"
- **The Ice Cream Test**: If we banned ice cream, would drownings stop?
- **Discovery**: Intervening breaks the correlation and reveals the hidden variable: **hot weather**

Rung 3: Counterfactuals

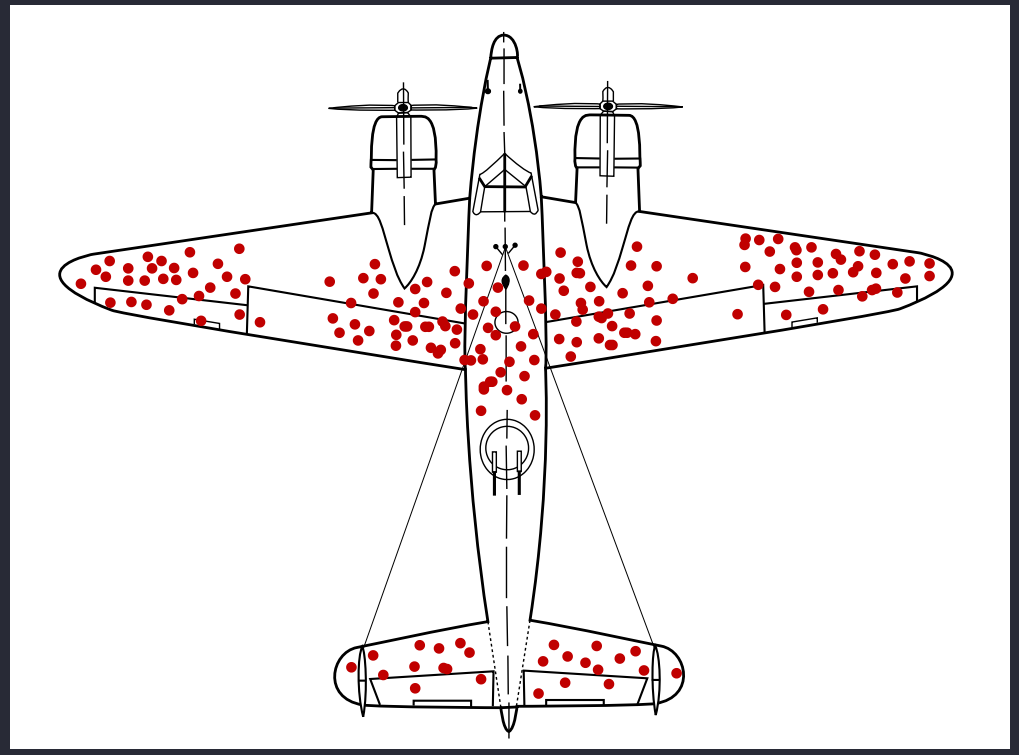
- **Nature**: The highest form of causal thinking; imagining things that didn't happen
- **Question**: "What if we had acted differently?"
- **Humanity**: This level allows us to explore alternate versions of the past to understand the present

Case Study: WWII Bomber Planes

- **The Problem**: Allied planes were being shot down; armor was needed but could only be placed in specific areas due to weight
- **Rung 1 Analysis**: Engineers mapped bullet holes on returning planes and wanted to armor the wings and tail because that's where the hits were seen

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Bomber bullet holes map



“ *credit: By Martin Grandjean (vector), McGeddon (picture), US Air Force (hit plot concept) - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=102017718* ”

Abraham Wald's Counterfactual

- **The Rung 3 Question**: Wald asked, "Where are the **missing** bullet holes?"
- **Survivor Bias**: The data only showed planes that made it back
- **The Genius Insight**: Any plane hit in the engines or cockpit never returned to be counted
- **The Solution**: Armor the places where you see *no* damage on survivors



Key Takeaways

- **Beyond the Data**: Real wisdom requires looking at information the data *isn't* telling you
- **Discussion Questions**:
  - What are the "missing bullet holes" in your own life or studies?
  - Whose stories are you not hearing at all?

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References

Topic	Source
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